LELME2150 - Thermal cycles

Homework 1 - Basic cycles

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Introduction

The objective of this homework is to become familiar with the numerical modelling of fundamental thermal cycles: the Brayton cycle and the Rankine cycle. Through the exercises, students will be able to:

- Review the basic thermodynamic cycles;
- Learn to use numerical thermodynamic tables (i.e. `CoolProp`, one of the most popular open source libraries);
- Understand which simplifying assumptions made in the LMECA1855 course can be easily avoided. In this homework, we get rid of the constant c_p assumption. Figures 1 and 2 show that c_p varies significantly with temperature, whereas its dependence on pressure is much weaker.
- Become familiar with parametric studies of thermal cycles and discover the beginnings of optimisation.

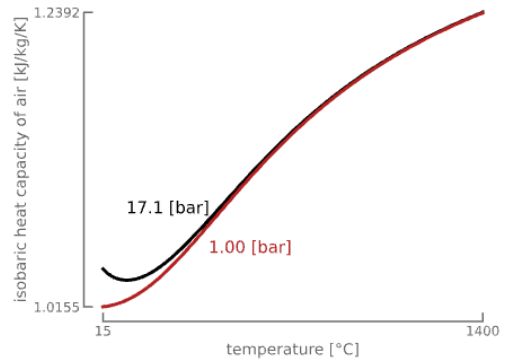


Figure 1: Isobaric heat capacity of air in function of temperature for two different pressures.

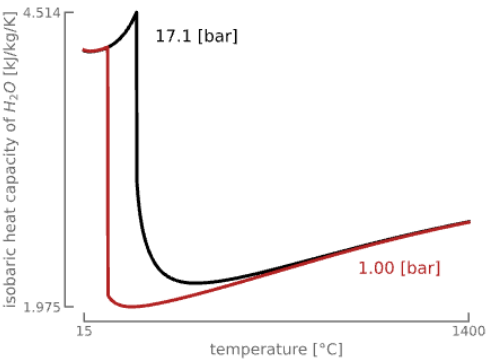


Figure 2: Isobaric heat capacity of water in function of temperature for two different pressures

In this course, the language used will be `Python` and the thermodynamic properties will be obtained via `CoolProp`.

This homework will not be graded, but you are strongly encouraged to complete it carefully, as it will help you prepare for the following homeworks and the project. Each group that submits a valid homework will receive feedback.
The **deadline** for this homework is **Sunday 20 September at 23:59**. Before submitting, we strongly recommend testing your code. A code that does not pass the basic test provided will not receive any feedback. Please submit only your `Python` scripts (two files) in the designated area on Moodle.

Preliminary tasks

1. Find a groupmate and select a group on Moodle;
2. Download the signatures (brayton_cycle_group_xx.py and rankine_cycle_group_xx.py);
3. Rename the scripts with your group number (0x if number lower than 10);
4. Download the test code (basic_cycles_test.py);
5. Install `CoolProp` (see this guide).

Exercice 1 - Brayton cycle

Consider the gas turbine cycle as shown in Figure 3.

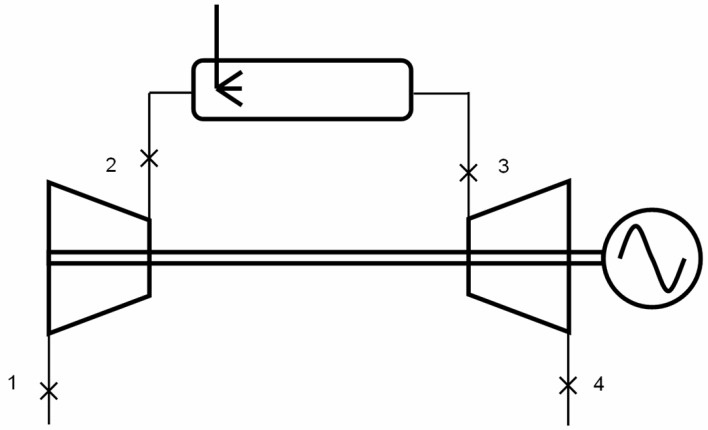


Figure 3: Basic gas turbine cycle.

In LMECA1855, we assumed a constant c_p for air. However, this is far from being a realistic assumption. Indeed, the c_p of gaseous fluids is strongly related to temperature and pressure. In this course, we will therefore consider it as variable. Using `CoolProp`, you can verify this:

```
pip install CoolProp
from CoolProp.CoolProp import PropsSI
PropsSI('CPMASS','P',1e+5,'T',300,'O2')
```

This example gives you the specific heat capacity of oxygen at 1 bar and 300 K.

In this exercise, you will have to simulate a gas turbine (i.e. Brayton cycle) using air as the working fluid (e.g. externally fired gas turbine, ...). The problem assumptions are as follows:

- The change in composition between the air and the combusted gases is neglected;
- The addition of mass during combustion is also neglected;
- The default molar air composition is 21% of O_2 and the rest of N_2;

- The transformations in the turbine and the compressor are characterized with the polytropic model;
- The pressure losses in the combustion chamber are taken into account;
- The mechanical efficiencies of the turbine and compressor are taken into account.

The problem data can be found in the test code; however, they are strictly indicative. You can indeed vary them to study their effect on the cycle. You can also refer to book Chapter 3 for more info.

Your goal is to implement your model in the evaluate function of brayton_cycle_group_xx.py. To do so, compute the thermodynamic state variables $p, T, s - s_1, h - h_1$ throughout the cycle (hint: to compute the entropy and enthalpy difference, the c_p must be integrated).

When running the test code with the provided data, you should obtain the same graphs as in Figure 4.

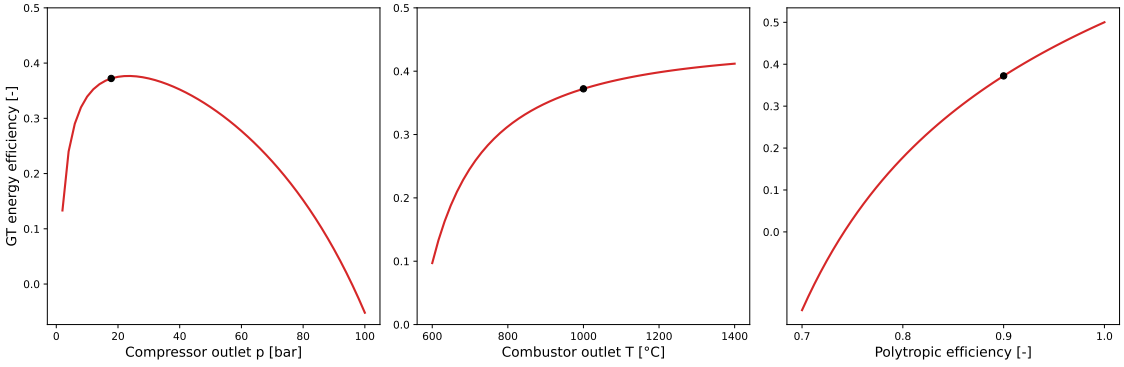


Figure 4: Energetic efficiency of the Brayton cycle

Exercice 2 - Rankine cycle

Consider the steam turbine cycle as shown in Figure 5. It is a closed-cycle consisting of 4 elements: a steam generator, an expansion turbine, a condenser and a pump.
In this exercise, you will simulate a steam turbine (i.e. Rankine cycle) using pure working fluids (H_2O as default). The problem assumptions are as follows:

- The steam is super-heated at the turbine inlet;
- Adiabatic expansion in the turbine down to the saturation pressure, the isentropic efficiency of the turbine is taken into account;
- Isobaric condensation;
- Sub-cooling of 3 K at the pump inlet (state 1);
- The efficiency of the steam generator is taken into account;
- The mechanical efficiency of the turbine is taken into account;

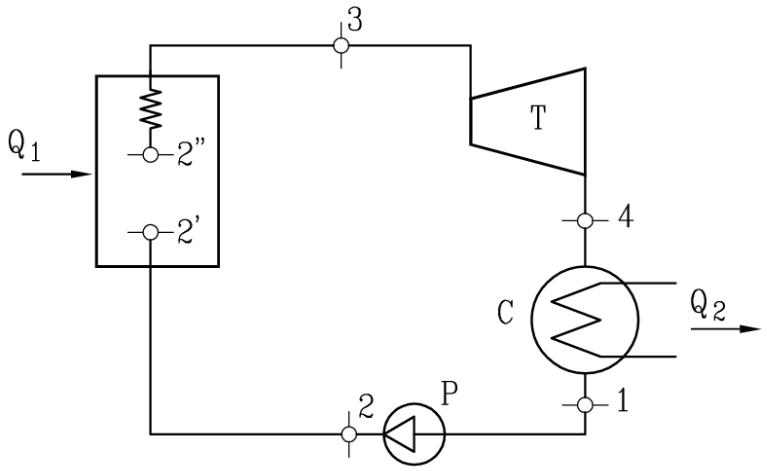


Figure 5: Basic steam turbine cycle.

- The pump is represented with the internal efficiency model;
- The pressure losses in the steam generator are taken into account;

The problem data can be found in the test code; however, they are strictly indicative. You can indeed vary them to study their effect on the cycle. You can also refer to book Chapter 2 for more info (you can compare your results to pp 62).

Your goal is to implement your model in the evaluate function of rankine_cycle_group_xx.py. To do so, compute the states p, T, s, h, x of the fluid during the cycle.

When running the test code with the provided data, you should obtain the same graphs as in Figure 6.

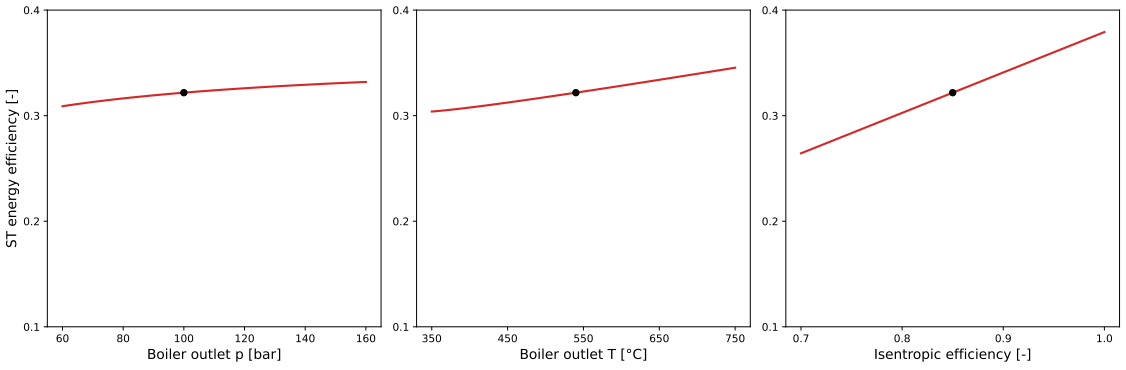


Figure 6: Energetic efficiency of the Rankine cycle

Going further? Instead of H2O, try to use propane as working fluid with $T_3 = 373.15$ K (i.e. Organic Rankine Cycle, ORC). What is the pressure level that maximises the efficiency?